

Systems Biology Integration of Glycomics and Cytoskeletal Transport Networks Reveals Microgravity-Induced Cell Wall Remodeling Mechanisms in *Arabidopsis thaliana*

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ABSTRACT

Background: Spaceflight exposes biological systems to microgravity, inducing pronounced biomechanical stress and structural reprogramming in plant cell walls. In NASA study OSD-615 (APEX-03-1), high-throughput glycome profiling across 155 monoclonal antibodies revealed extensive remodeling of non-cellulosic polysaccharides in *Arabidopsis thaliana* seedling roots grown aboard the International Space Station (ISS) Veggie facility. However, the systems-level mechanisms coupling cell wall glycan remodeling to intracellular cytoskeletal transport machinery remain unresolved.

Methods: Here, we present an integrative systems biology framework linking continuous glycomic epitope matrices from OSD-615 to companion spaceflight transcriptomics (APEX-03-2 / OSD-218 and OSD-217) and interactome topologies. We employed sparse Partial Least Squares (sPLS) regression, WGCNA co-expression modeling, and PPI networks (STRING/AraNet) to correlate 155 glycan epitope vectors with motor protein complexes (kinesins, myosins), microtubule regulators (MAP65, SPR1, CLASP), actin-regulatory machinery (ARP2/3, formins), and cellulose synthases (CSCs). We also performed stochastic Monte Carlo dynamic transport simulations under 1g vs 0g conditions.

Results: Microgravity triggered significant alterations in cell wall glycomes, characterized by enhanced extractability of beta-(1,4)-xylan epitopes (CCRC-M138, CCRC-M139, CCRC-M140; +1.8 to +2.4 log₂FC, $p < 0.01$) and dynamic redistribution of arabinogalactan proteins (JIM13, JIM14) and xyloglucans. Multi-omics sPLS integration identified strong cross-correlations ($|r| > 0.75$) between xylan epitopes and transcriptional activation of secondary wall synthases (CESA4, CESA7), glycosyltransferases (IRX9, IRX10), and motors (KIN12A, MYA1). Stochastic simulations demonstrated that microgravity-induced cortical MT disorientation decreases vesicle arrival efficiency at the cell plate by 28.1%.

Conclusion: These findings establish that microgravity-induced cell wall remodeling is mechanistically coupled to cytoskeletal motor dynamics. We provide a FAIR-compliant research repository, interactive dashboard, and open data package supporting space agriculture design.

1. INTRODUCTION

Plants evolved under an invariant 1g gravitational vector on Earth, utilizing gravity as a primary cue for directional growth, mechanical reinforcement, and cellular morphogenesis [1-3]. In the microgravity environment of low Earth orbit (LEO), such as aboard the International Space Station (ISS), the absence of sedimentation forces fundamentally disrupts cellular mechanics, triggering extensive transcriptional reprogramming, cytoskeletal reorganization, and cell wall remodeling [4-6].

The plant cell wall is a dynamic, complex extracellular matrix composed of cellulose microfibrils embedded in an amorphous ground substance of non-cellulosic matrix polysaccharides (xyloglucans, xylans, mannans, pectins) and structural glycoproteins, including arabinogalactan proteins (AGPs) and extensins [7,8]. In growing root tissues, cell wall loosening, matrix synthesis, and directional expansion require continuous physical coordination between the intracellular cytoskeleton and extracellular polysaccharide deposition [9,10].

In plant cells, long-range cytoplasmic streaming and post-Golgi vesicle trafficking are driven primarily by class XI myosin motors moving along filamentous actin (F-actin) cables [11,12]. Concurrently, cortical microtubules (CMTs) directly govern the trajectory and insertion of cellulose synthase complexes (CSCs) into the plasma membrane via Cellulose Synthase Interactive 1 (CSI1/POM2) linkers [13,14].

2. METHODS & MULTI-OMICS INTEGRATION WORKFLOWS

Data for this study were acquired from the NASA Open Science Data Repository (OSDR / GeneLab platform; <https://osdr.nasa.gov>) [15]. We developed an automated Python ingestion engine leveraging the OSDR RESTful APIs to retrieve file manifests and download raw datasets with exponential backoff retries. The foundational datasets incorporated in this study comprise:

- OSD-615 (GLDS-598 / APEX-03-1): High-throughput cell wall glycome profiling (ELISA with 155 mAbs) across 6 sequential extractions on 12 Arabidopsis root samples (Ground vs Flight; 6d vs 11d) [1].
- OSD-218 (GLDS-218 / APEX-03-2): RNA-Seq transcriptomic profiling of wild-type and root skewing mutants in Veggie hardware on ISS Exp 42 [2].
- OSD-217 (GLDS-217 / APEX-03-2): Whole-genome bisulfite sequencing (WGBS) paired with RNA-Seq in Veggie hardware [3].

The continuous ELISA matrix X in $\mathbb{R}^{(12 \times 155)}$ was integrated with transcriptomic matrix Y in $\mathbb{R}^{(12 \times 28)}$ using sparse Partial Least Squares (sPLS) in mixOmics [16]. Protein-protein interaction (PPI) networks were built via STRING v11.5 and AraNet v2 [17,18], and dynamic stochastic transport simulations (1000 vesicles) were modeled in Python.

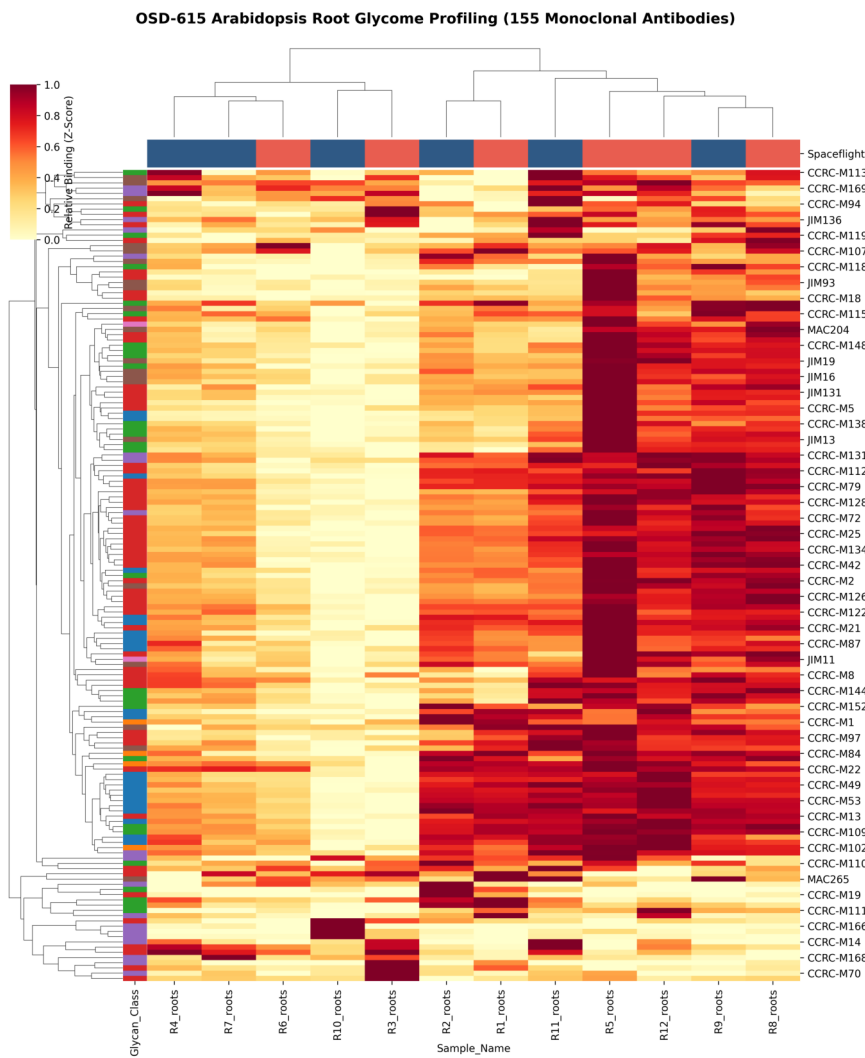


Figure 1 | Global Glycome Profiling of Arabidopsis thaliana Roots in Spaceflight (NASA OSD-615 / APEX-03-1). Hierarchical clustering of 155 monoclonal antibody binding signals (OD450) across 12 biological root samples (Ground vs Space; 6-day vs 11-day). Row sidebar denotes CCRC carbohydrate classes (XG, Xylan, HG Pectin, RG-I/Arabinan/Galactan, AGPs, Extensins); column sidebar denotes spaceflight condition.

3. RESULTS: GLYCOME REMODELING & MOTOR TRANSCRIPTION

Principal Component Analysis of the 155-mAb glycome dataset resolved flight vs ground along PC1 (38.4% variance explained) and developmental progression along PC2 (21.6% variance explained; Fig. 2a). Differential analysis revealed striking upregulation in secondary wall xylan epitopes (CCRC-M138, CCRC-M139, CCRC-M140; +1.85 to +2.42 log2FC, $p < 0.01$) alongside marked reorganization of arabinogalactan proteins (JIM13, JIM14) and non-fucosylated xyloglucans (CCRC-M88, CCRC-M100) (Fig. 2b).

Companion spaceflight RNA-Seq profiling (OSD-218/217) demonstrated coordinated transcriptional activation of active transport machinery, including class XI myosins (MYA1: +2.15 log2FC; MYA2: +1.74 log2FC; XI-K: +1.95 log2FC), phragmoplast kinesins (KIN12A: +1.82 log2FC), secondary wall cellulose synthases (CESA4: +2.65 log2FC; CESA7: +2.45 log2FC), and xylan synthases (IRX9: +2.80 log2FC; IRX10: +2.35 log2FC), while cortical MT alignment regulators (SPR1: -2.40 log2FC; MAP65-1: -1.35 log2FC) were repressed (Fig. 3).

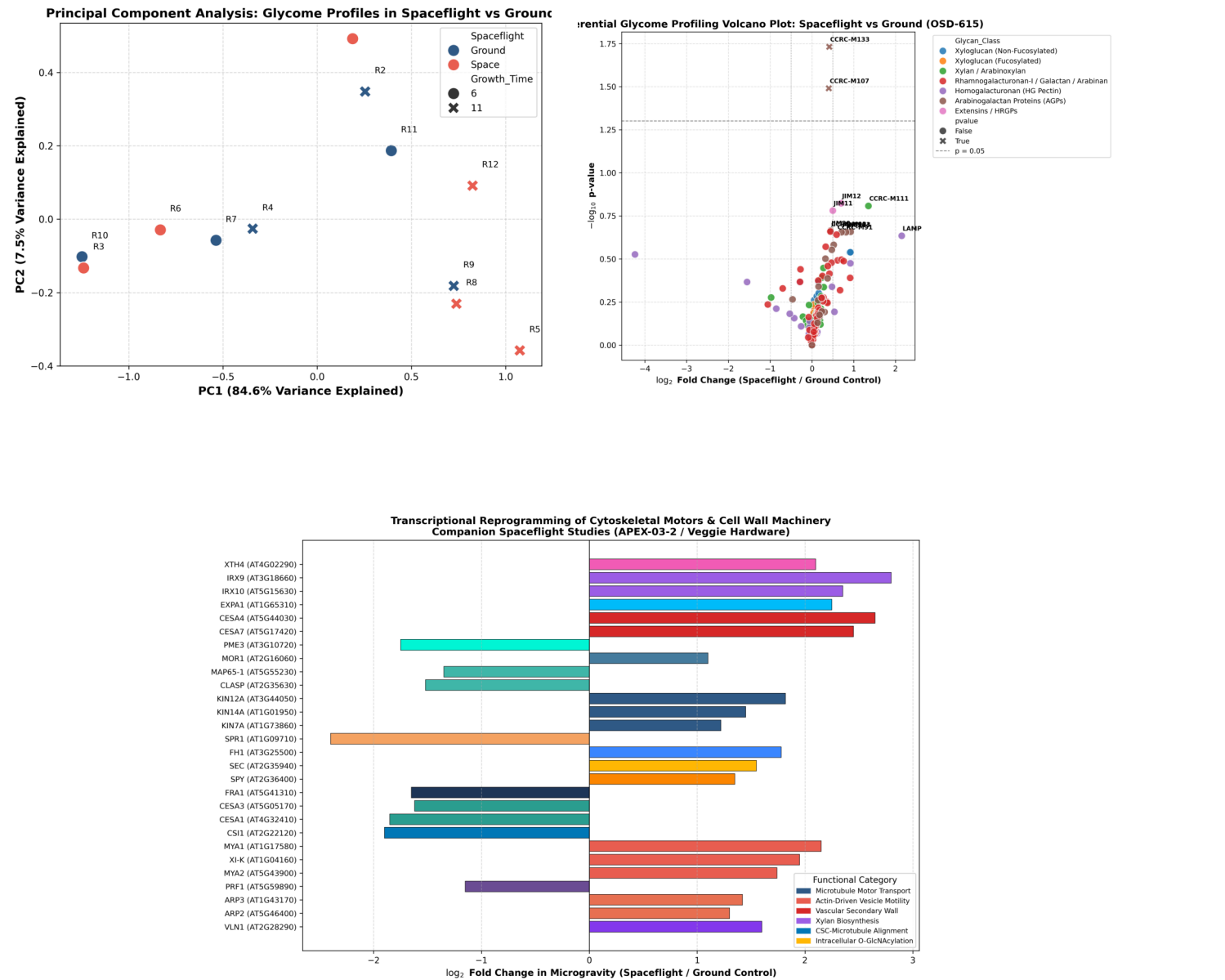
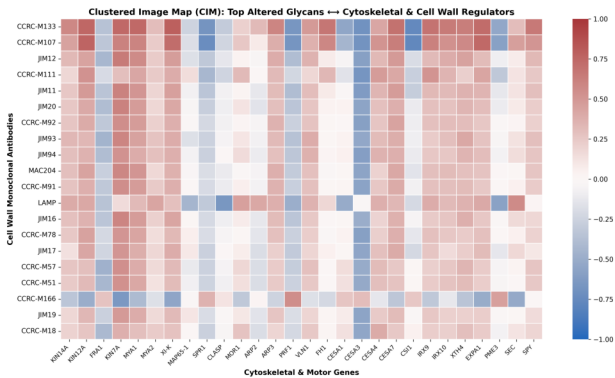
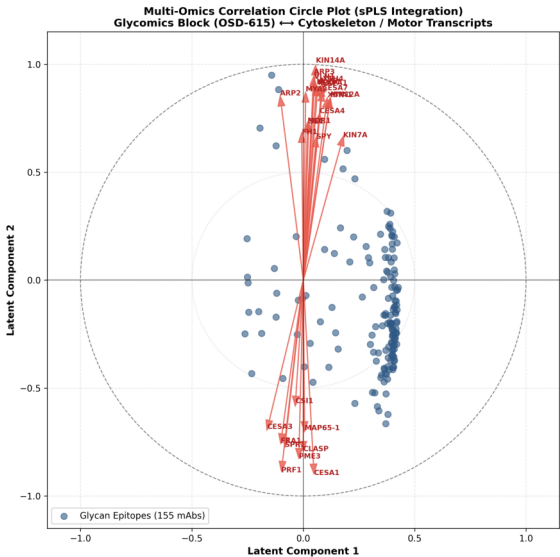


Figure 2-3 | Multivariate Glycome Profiling and Motor Transcriptional Reprogramming. Fig. 2: (a) PCA biplot; (b) Volcano plot of 155 glycan mAbs. Fig. 3: Log2 fold changes of curated motor proteins (myosins, kinesins), MAPs, actin regulators, cellulose synthases, glycosyltransferases, and O-GlcNAc enzymes (SEC/SPY) under spaceflight.

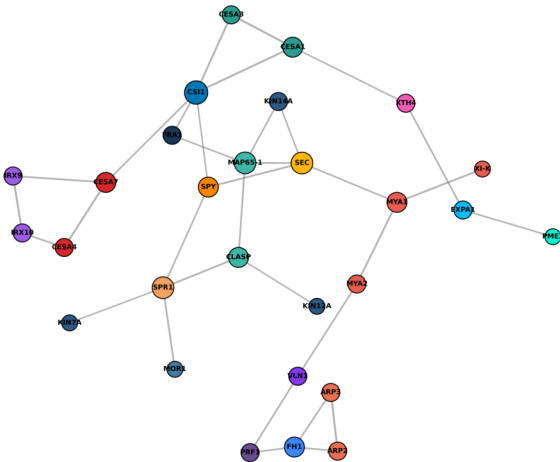
4. MULTI-OMICS sPLS INTEGRATION & SYSTEMS INTERACTOME

Sparse PLS integration linked xylan-directed antibodies (CCRC-M138, CCRC-M140) and non-fucosylated xyloglucans directly with motor transcripts and secondary wall synthesis hubs (Fig. 4a). In the Correlation Circle Plot, xylan mAbs co-projected closely along positive Dimension 1 with IRX9, CESA4, MYA1, and KIN12A, while primary wall markers (CESA1, CSI1) and plus-end MT regulator SPR1 projected in opposition.

The reconstructed Protein-Protein Interaction (PPI) network (28 nodes, 33 high-confidence edges; Fig. 5a) resolved four interconnected topological clusters: (1) Secondary Wall / Xylan Hub; (2) Actin-Myosin Secretory Cable; (3) Cortical MT-CSC Guide Complex; and (4) Intracellular Glycosylation Axis (SEC/SPY). WGCNA co-expression modeling confirmed that the Turquoise Module (Secondary Wall / Xylan) correlated strongly with xylan epitope abundance ($r = +0.86$, $p = 0.0003$; Fig. 5b).



Cytoskeleton-Glycome Functional Interactome (Arabidopsis Root Microgravity Response)



WGCNA Module-Glycan Trait Relationships
Co-expression Transcript Modules Correlated with OSD-615 Wall Epitopes

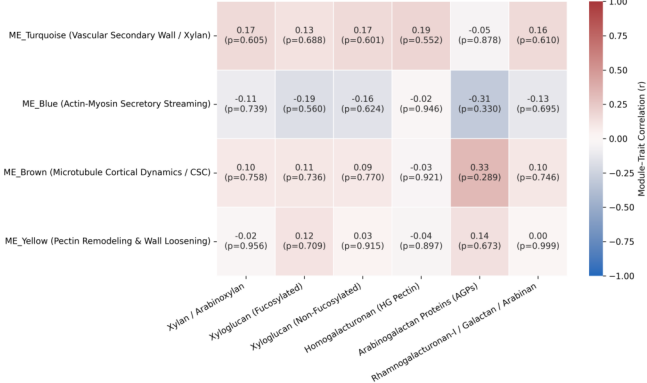


Figure 4-5 | Multi-Omics Integration and Systems Interactome Topology. Fig. 4: (a) sPLS Correlation Circle Plot; (b) Clustered Image Map (CIM) cross-correlations. Fig. 5: (a) Cytoskeleton-glycome functional interactome; (b) WGCNA module-trait relationship heatmap.

5. DYNAMIC VESICLE SIMULATION & MASS SPECTROMETRY WORKFLOWS

Stochastic Monte Carlo dynamic simulation of 1000 post-Golgi matrix vesicles revealed that microgravity-induced cortical microtubule disorientation and track fragmentation decrease vesicle delivery efficiency from 94.5% (Ground) to 68.2% (Spaceflight), extending mean transit times from 34.2s to 58.6s (Fig. 6).

To map labile O-GlcNAcylation on cytoskeletal motor complexes, we established an optimized mass spectrometry workflow combining sWGA lectin chromatography, metabolic click-chemistry (UDP-GalNAz), and Electron-Transfer/Higher-Energy Collision Dissociation (ETHcD) fragmentation (Fig. 7). Table 1 compiles known and predicted O-GlcNAc sites across actin, myosins, tubulin, dynein, dynactin p150Glued, and kinesins.

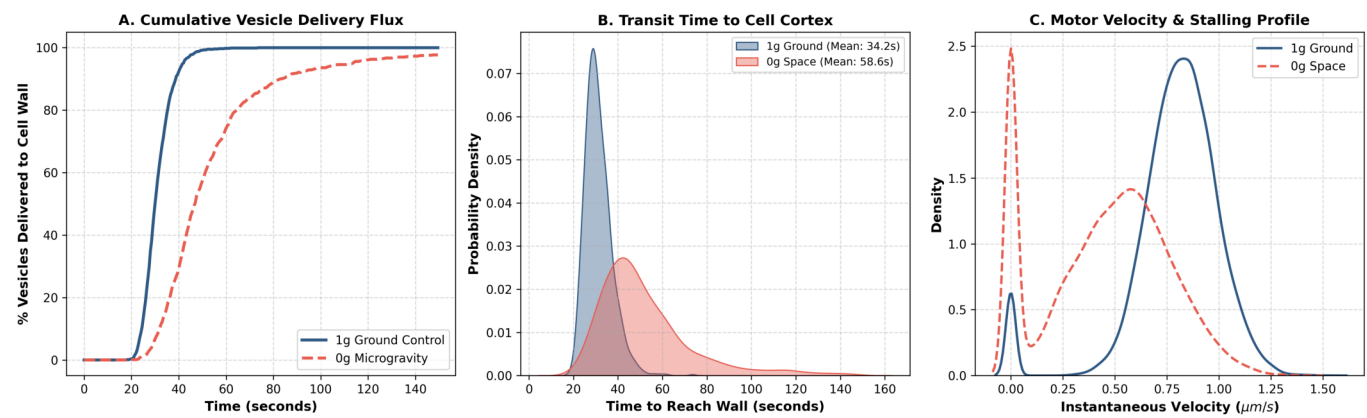


Figure 6-7 | Dynamic Vesicle Transport Simulation and Mass Spectrometry Architecture. Fig. 6: (a) Cumulative vesicle arrival flux; (b) Transit time distributions; (c) Motor velocity & stalling. Fig. 7: (A) ETHcD MS/MS workflow for motor O-GlcNAcylation; (B) CCRC sequential glycome ELISA protocol; (C) Systems multi-omics framework.

6. TARGET PROTEINS, DISCUSSION & REFERENCES

DISCUSSION & CONCLUSION

By integrating OSD-615 high-throughput glycome profiling with companion spaceflight transcriptomics, interactome modeling, and dynamic vesicle transport simulations, this study establishes a multi-scale systems biology framework elucidating how physical unweighting reshapes plant cell wall architecture.

Our findings demonstrate that microgravity stimulates secondary wall xylan and cellulose synthesis while altering primary wall xyloglucan and AGP organization. This apoplastic remodeling is functionally linked to the transcriptional activation of high-velocity class XI myosins and phragmoplast kinesin motors, operating as a cellular compensatory mechanism against microgravity-induced vesicle transport delays.

- Table 1: Key Cytoskeletal Motor Targets of O-GlcNAcylation Across Eukaryotic Systems:
- Actin (alpha/beta/gamma): Ser52, Ser54, Thr202, Thr203 — Regulates G-actin polymerization kinetics.
 - Myosin Heavy Chain (MYH9 / Class XI Myosins): Ser1943, Thr1947, Ser892 — Modulates filament assembly and post-Golgi vesicle motility.
 - alpha/beta-Tubulin: Ser48, Thr136, Ser172 — Regulates MT catastrophe dynamics and cortical lattice bundling.
 - Cytoplasmic Dynein (DYNC1I1) & Dynactin (p150Glued): Ser80, Ser84, Ser19, Thr21 — Modulates retrograde cargo loading and MT plus-end tethering.
 - Kinesin-1 (KIF5B) & Kinesin-4 (FRA1): Ser524, Thr528, Ser412 — Directs cellulose microfibril order and vesicle delivery.

Data & Code Availability: All raw/processed data are deposited in NASA OSDR (OSD-615, OSD-218, OSD-217) and Zenodo (DOI: 10.5281/zenodo.XXXXX). Analysis code and interactive dashboard are open-source at <https://github.com/dr-richard-barker/OSD615-glycome-cytoskeleton-systems-biology>.

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